# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

| | | | |
|---|---|---|---|
| **Course Code:** | CSA17 | **Course Title:** | Artificial Intelligence |
| **CO Assessed:** | CO1 – Apply (BL3) | **Assessment:** | Assessment Tool 1 – Analytical Problem Solving |
| **Weightage:** | 35% of CIA | **Duration / Marks:** | 60 mins |
| **CO1 Statement:** | Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3) | | |
| **Student Name:** | G.SANGHAMITRA | **Reg. No.:** | 1924260351 |
| **Section / Batch:** | D/ 2026 | **Date:** | 03-10-2026 |

**Code of Conduct**

*I G.Sanghamitra,1924260351(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Sanghamitra

**Instructions**

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

| Section / Topic | Focus | Question Nos. |
|---|---|---|
| **1. Foundations and History of AI** | Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI | **Q3** |
| **2. Intelligent Agents: Agents and Environments** | Agent function and program, PEAS description, task environment properties, agent architectures | **Q1, Q2** |

| | | |
|---|---|---|
| **3. Concept of Rationality** | Performance measure, percept sequence, expected utility, rational versus omniscient agent | **Q1, Q4** |

**Annexure A – Analytical Problem Solving**

## Question 1: Two-Room Vacuum World

*Topic: Agents and environments, concept of rationality | 15 minutes*

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

**Performance measure over a lifetime of T = 6 time steps:** +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

**(a)** Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

**(b)** Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

**(c)** Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

**(d)** Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

**(e)** Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

# Solution:

**a) PEAS Description — 4 Marks**

| PEAS | Description |
|---|---|
| **Performance Measure (P)** | +1 for every clean square at the end of each time step; −p for every Left/Right movement |
| **Environment (E)** | Two squares **A and B**; initially both are Dirty; dirt never reappears |
| **Actuators (A)** | **Suck, Left, Right, NoOp** |
| **Sensors (S)** | Perceives its **current location** (A or B) and whether the square is **Dirty or Clean** |

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**(b) Agent X — Simple Reflex Agent**

**Initial state:** A = Dirty, B = Dirty
**p = 1**

Agent X rule:

- Dirty → **Suck**
- Clean → move to the other square

| Step | Location | Percept | Action | Clean squares at end | Move penalty | Step score |
|---|---|---|---|---|---|---|
| 1 | A | Dirty | **Suck** | 1 | 0 | **1** |
| 2 | A | Clean | **Right** | 1 | −1 | **0** |
| 3 | B | Dirty | **Suck** | 2 | 0 | **2** |
| 4 | B | Clean | **Left** | 2 | −1 | **1** |
| 5 | A | Clean | **Right** | 2 | −1 | **1** |
| 6 | B | Clean | **Left** | 2 | −1 | **1** |

**Total score of X**

$$ 1+0+2+1+1+1=\boxed{6} $$

So,

$$ \boxed{X=6\text{ for }p=1} $$

The agent keeps moving because it has **no memory** that both squares are already clean.

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**(c) Agent Y — Reflex Agent with Internal State**

Agent Y remembers whether **both squares have been cleaned**.

Rules:

- Dirty → **Suck**
- Clean + both not cleaned → move to other square
- Clean + both cleaned → **NoOp**

| Step | Location | Percept | Action | Clean squares at end | Move penalty | Step score |
|---|---|---|---|---|---|---|
| 1 | A | Dirty | **Suck** | 1 | 0 | **1** |
| 2 | A | Clean | **Right** | 1 | −1 | **0** |
| 3 | B | Dirty | **Suck** | 2 | 0 | **2** |
| 4 | B | Clean | **NoOp** | 2 | 0 | **2** |
| 5 | B | Clean | **NoOp** | 2 | 0 | **2** |
| 6 | B | Clean | **NoOp** | 2 | 0 | **2** |

**Total score of Y**

$$ 1+0+2+2+2+2=\boxed{9} $$

Therefore,

$$ \boxed{Y=9\text{ for }p=1} $$

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**(d) Score as a function of p — 3 Marks**

**Agent X**

Agent X makes **4 moves**:

- Step 2: Right
- Step 4: Left
- Step 5: Right
- Step 6: Left

Total clean-square reward:

$$ 1+1+2+2+2+2=10 $$

Total movement penalty:

$$ 4p $$

Therefore,

$$ \boxed{X(p)=10-4p} $$

**Agent Y**

Agent Y makes only **1 move**:

- Step 2: Right

Total clean-square reward:

$$ 1+1+2+2+2+2=10 $$

Movement penalty:

$$ p $$

Therefore,

$$ \boxed{Y(p)=10-p} $$

**When is Y strictly better?**

$$ Y(p)>X(p) $$$$ 10-p>10-4p $$$$ -p>-4p $$$$ 3p>0 $$

Therefore,

$$ \boxed{p>0} $$

So **Y is strictly better than X whenever the movement penalty is positive**.

For $p=0$, both have the same score:

$$ X(0)=Y(0)=10 $$

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**(e) Which Agent is Rational? — 4 Marks**

**Agent Y is rational for $p>0$)** because it uses its percept history/internal state to remember that both squares have already been cleaned. Once everything is clean, it chooses **NoOp**, avoiding unnecessary movements and their penalties.

Agent X is a **simple reflex agent**. It only considers the **current percept**. When it sees a clean square, it always moves to the other square. Therefore, after both squares are clean, X continues moving:

$$ A\rightarrow B\rightarrow A\rightarrow B\rightarrow A $$

and receives unnecessary movement penalties.

**Important point about rationality**

Rationality depends on:

1. **Performance measure** — here, keeping squares clean gives reward and movement gives a penalty.
2. **Percept history/internal state** — Y remembers what happened previously and knows that both squares are clean.
3. **Available actions** — Y can choose **NoOp** once cleaning is complete.

Although X and Y receive the **same percepts at each location**, Y can use its **internal state/history**, while X cannot.

Therefore, for $p>0$:

$$ \boxed{\text{Y is rational and achieves a higher performance score than X.}} $$

**Final answers**

$$ \boxed{X(p)=10-4p} $$$$ \boxed{Y(p)=10-p} $$$$ \boxed{Y>X\text{ when }p>0} $$

For $p=1$:

$$ \boxed{X=6,\quad Y=9} $$

## Question 2: Task Environment Analysis and Agent Selection

*Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes* Three

AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

**Difficulty score convention:** for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

| Property | Easier value (0) | Harder value (1) |
|---|---|---|
| **Observability** | Fully observable | Partially observable |
| **Number of agents** | Single agent | Multi-agent |
| **Determinism** | Deterministic | Stochastic |
| **Episodic / Sequential** | Episodic | Sequential |
| **Static / Dynamic** | Static | Dynamic (semi-dynamic = 0.5) |
| **Discrete / Continuous** | Discrete | Continuous |

**(a)** Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

**(b)** Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**

**(c)** For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

**(d)** Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

# Solution:

**a) Classification of E1, E2 and E3 — 9 Marks**

| Property | E1: Autonomous Taxi | E2: Part-Picking Robot | E3: Chess Agent |
|---|---|---|---|
| **Observability** | **Partially observable** — sensors cannot see every road condition, blind spots, etc. | **Partially observable** — camera may miss or misclassify defective parts | **Fully observable** — complete chessboard position is visible |
| **Number of agents** | **Multi-agent** — other drivers, pedestrians and vehicles interact | **Single agent** — robot performs the picking task independently | **Multi-agent** — two players compete |
| **Determinism** | **Stochastic** — traffic and other drivers are unpredictable | **Stochastic** — moving parts and vision errors introduce uncertainty | **Deterministic** — a legal chess move has a predictable board result |
| **Episodic / Sequential** | **Sequential** — current driving decisions affect future states | **Sequential** — timing and previous actions affect subsequent picking | **Sequential** — each move affects future game positions |
| **Static / Dynamic** | **Dynamic** — traffic and surroundings change continuously | **Dynamic** — conveyor belt and parts are continuously moving | **Semi-dynamic** — board does not change by itself, but the clock changes during thinking |

| Property | E1: Autonomous Taxi | E2: Part-Picking Robot | E3: Chess Agent |
|---|---|---|---|
| **Discrete / Continuous** | **Continuous** — position, speed, steering and time vary continuously | **Continuous** — belt motion, robot position and timing are continuous | **Discrete** — finite board, pieces and legal moves |

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**(b) Difficulty Score — 4 Marks**

According to the given convention:

- Harder property = **1**
- Easier property = **0**
- Semi-dynamic = **0.5**

**E1 — Autonomous Taxi**

| Property | Classification | Score |
|---|---|---|
| Observability | Partially observable | 1 |
| Agents | Multi-agent | 1 |
| Determinism | Stochastic | 1 |
| Episodic/Sequential | Sequential | 1 |
| Static/Dynamic | Dynamic | 1 |
| Discrete/Continuous | Continuous | 1 |
| **Total** | | **6** |

$$\boxed{E1=6}$$

**E2 — Part-Picking Robot**

| Property | Classification | Score |
|---|---|---|
| Observability | Partially observable | 1 |
| Agents | Single agent | 0 |
| Determinism | Stochastic | 1 |
| Episodic/Sequential | Sequential | 1 |
| Static/Dynamic | Dynamic | 1 |
| Discrete/Continuous | Continuous | 1 |
| **Total** | | **5** |

$$\boxed{E2=5}$$

**E3 — Chess Agent**

| Property | Classification | Score |
|---|---|---|
| Observability | Fully observable | 0 |
| Agents | Multi-agent | 1 |
| Determinism | Deterministic | 0 |
| Episodic/Sequential | Sequential | 1 |
| Static/Dynamic | Semi-dynamic | 0.5 |
| Discrete/Continuous | Discrete | 0 |
| **Total** | | **2.5** |

$$\boxed{E3=2.5}$$

**Ranking**

Based strictly on the given difficulty-score convention:

$$\boxed{E1\;(6) > E2\;(5) > E3\;(2.5)}$$

So:

**Hardest → E1 → E2 → E3 → Easiest**

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**(c) Agent Architecture Selection — 4 Marks**

**Hardest: E1 — Autonomous Taxi**

**Suitable architecture: Learning Agent**

A **learning agent** is suitable because the taxi operates in a:

- Partially observable environment
- Multi-agent environment
- Stochastic environment
- Sequential environment
- Dynamic environment
- Continuous environment

The taxi must deal with changing traffic, pedestrians, road conditions and other drivers. A learning agent can use past experience and sensor information to improve its decisions.

It can learn things such as:

- Traffic patterns
- Safe driving behaviour
- Route selection
- Behaviour of other road users

$$\boxed{\text{E1 → Learning Agent}}$$

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**Easiest: E3 — Chess**

**Suitable architecture: Goal-based Agent**

Chess has:

- Full observability
- Discrete states and actions
- Deterministic rules
- Sequential decision-making
- Two competing agents

The main goal is to **checkmate the opponent**. Therefore, a goal-based agent can evaluate possible moves and select moves that help achieve the goal.

A practical chess system may also use **utility-based** or **learning** techniques, but among the listed choices, a **goal-based agent** directly matches the basic task description.

\[ \boxed{\text{E3 → Goal-Based Agent}} \]

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**(d) Interpretation — 3 Marks**

The complexity of an agent program depends not only on **what task it performs**, but also on the **properties of its environment**.

For example, an autonomous taxi must handle **partial observability, multiple agents, uncertainty, sequential decisions, continuous changes and dynamic surroundings**. Therefore, it requires a much more complex architecture.

In contrast, chess is **fully observable, deterministic and discrete**, so the basic environment is easier to model, even though it still requires planning because it is sequential and multi-agent.

Therefore:

**Environment properties determine what information the agent needs, how much uncertainty it must handle, whether it needs memory, planning, learning and utility evaluation, and consequently how complex its agent program must be.**

**Final Summary**

| Environment | Score | Rank | Suitable Architecture |
|---|---|---|---|
| **E1 – Autonomous Taxi** | **6** | Hardest | **Learning Agent** |
| **E2 – Part-Picking Robot** | **5** | 2nd | — |
| **E3 – Chess Agent** | **2.5** | Easiest | **Goal-Based Agent** |

$$\boxed{\text{Hardest: E1} \quad\rightarrow\quad \text{E2}\quad\rightarrow\quad \text{Easiest: E3}}$$

## Question 3: Turing Test Data Across the History of AI

*Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes*

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

| System | Era / technique | Interrogations | Correctly identified |
|---|---|---|---|
| **ELIZA** | 1960s, keyword pattern matching | 50 | 47 |
| **Chatbot B** | 2010s, scripted rules plus statistical NLP | 50 | 36 |
| **System C** | 2020s, large neural language model | 50 | 29 |

**Reference:** Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

**(a)** Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

**(b)** Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

**(c)** Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

**(d)** Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

# Solution:

**(a) Correct-identification rate and fooling rate — 6 Marks**

There are **50 interrogations** for each system.

**Formula**

$$\text{Correct-identification rate} =\frac{\text{Correctly identified}}{50}\times100$$

Since the machine is correctly identified when the judge says **"machine"**, the fooling rate is:

$$\text{Fooling rate} =100-\text{Correct-identification rate}$$

| System | Correctly Identified | Correct-identification Rate | Fooling Rate |
|---|---|---|---|
| **ELIZA** | 47/50 | $(47/50\times100=)$ **94%** | **6%** |
| **Chatbot B** | 36/50 | $(36/50\times100=)$ **72%** | **28%** |
| **System C** | 29/50 | $(29/50\times100=)$ **58%** | **42%** |

Therefore:

$$\boxed{\text{ELIZA: 94\% identification, 6\% fooling}}$$$$\boxed{\text{Chatbot B: 72\% identification, 28\% fooling}}$$$$\boxed{\text{System C: 58\% identification, 42\% fooling}}$$

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**(b) Turing's 70% Criterion — 5 Marks**

Turing's reference says that after five minutes, an average interrogator would have **no more than a 70% chance of correctly identifying the machine**.

So, a system satisfies the criterion if:

$$\text{Correct-identification rate} \leq 70\%$$

**Checking each system**

| System | Correct Identification | Meets 70% Criterion? | Distance from 50% Guessing Level |
|---|---|---|---|
| **ELIZA** | 94% | ❌ No | (94-50=) **44 percentage points above** |
| **Chatbot B** | 72% | ❌ No | (72-50=) **22 percentage points above** |
| **System C** | 58% | ✅ Yes | (58-50=) **8 percentage points above** |

**Conclusion**

Only **System C** satisfies Turing's 70% criterion because judges correctly identified it only **58%** of the time, which is below 70%.

However, System C is still **8 percentage points above the 50% guessing level**, meaning judges performed somewhat better than pure guessing.

$$\boxed{\text{Only System C satisfies the 70\% criterion.}}$$

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**(c) Improvement in Fooling Rate — 5 Marks**

**1. ELIZA → System C**

ELIZA fooling rate:

$$6\%$$

System C fooling rate:

$$42\%$$

**Improvement in percentage points**

$$ 42-6=\boxed{36\text{ percentage points}} $$

**Improvement as a ratio**

$$ \frac{42}{6}=7 $$

Therefore, System C's fooling rate is:

$$ \boxed{7\times} $$

the fooling rate of ELIZA.

So:

$$ \boxed{\text{Improvement = 36 percentage points, or 7:1 ratio}} $$

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**2. Chatbot B → System C**

Chatbot B fooling rate:

$$ 28\% $$

System C fooling rate:

$$ 42\% $$

Therefore:

$$ 42-28=\boxed{14\text{ percentage points}} $$

So System C improved the fooling rate by:

$$ \boxed{14\text{ percentage points}} $$

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**(d) Interpretation — 4 Marks**

The **Turing Test measures "acting humanly"**, not necessarily genuine thinking or rationality.

System C has a **42% fooling rate**, meaning that in 42% of the interrogations, judges incorrectly thought the machine was human. This shows that System C can produce conversational behaviour that appears human-like under the conditions of the test.

However, this does **not prove that System C thinks like a human or possesses human-like understanding**.

**Chinese Room argument**

The **Chinese Room argument**, proposed by John Searle, illustrates this distinction. A system could manipulate symbols and produce appropriate responses according to rules without actually understanding the meaning of those symbols.

Similarly, System C may generate convincing language responses without this test establishing that it has:

- Human-like understanding
- Consciousness
- Genuine reasoning
- Human-like intentions

Therefore:

**Passing or performing well in a Turing-test-style conversation demonstrates successful human-like behaviour, but it does not by itself prove genuine thinking, understanding, or rationality.**

**Final Answers at a Glance**

| System | Identification | Fooling | 70% Criterion |
|---|---|---|---|
| **ELIZA** | 94% | 6% | ❌ |
| **Chatbot B** | 72% | 28% | ❌ |
| **System C** | 58% | 42% | ✅ |

\[ \boxed{\text{ELIZA → C improvement}=36\text{ points}=7\times} \]\[ \boxed{\text{Chatbot B → C improvement}=14\text{ points}} \]

## Question 4: Rational Route Choice under Uncertainty

*Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes*

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

**$U = 100 - T - 20 \cdot L$**

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

| Route | Normal time (min) | Probability of disruption | Time if disrupted (min) |
|---|---|---|---|
| **R1: Highway** | 30 | 0.30 (accident) | 70 |
| **R2: City road** | 45 | 0.10 (signal failure) | 60 |
| **R3: Ring road** | 50 | 0 (never disrupted) | 50 |

**(a)** Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

**(b)** Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

**(c)** A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

**(d)** Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

**(e)** Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

# Solution:

Given:
$$U=100-T-20L$$
where:

- $T$ = travel time in minutes
- $L=1$ if arrival is after 55 minutes
- $L=0$ if arrival is within 55 minutes
- Deadline = 55 minutes

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(a) Utility of each route in each outcome — 4 Marks
Route R1: Highway
Normal condition
Travel time:
$$T=30$$
Since $30<55$,
$$L=0$$
Therefore:
$$U=100-30-20(0) \boxed{U=70}$$
Disrupted — Accident
Travel time:
$$T=70$$
Since $70>55$, the taxi is late.
$$L=1$$
Therefore:
$$U=100-70-20(1) \boxed{U=10}$$

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Route R2: City Road
Normal condition
$$T=45,\quad L=0 \quad U=100-45-20(0) \boxed{U=55}$$
Disrupted — Signal Failure
$$T=60$$
Since $60>55$, the taxi is late.
$$L=1 \quad U=100-60-20 \boxed{U=20}$$

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Route R3: Ring Road
R3 is never disrupted.

$$ T=50 $$
Since $50<55$,
$$ L=0 $$
Therefore:
$$ U=100-50 $$$$ \boxed{U=50} $$
Summary

| Route | Outcome | Time $(T)$ | Late $(L)$ | Utility |
|---|---|---|---|---|
| R1 | Normal | 30 | 0 | 70 |
| R1 | Accident | 70 | 1 | 10 |
| R2 | Normal | 45 | 0 | 55 |
| R2 | Signal failure | 60 | 1 | 20 |
| R3 | Normal | 50 | 0 | 50 |

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(b) Expected Utility and Rational Choice — 5 Marks
Expected utility is:
$$ EU=\sum P(\text{outcome})\times U(\text{outcome}) $$

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R1 — Highway
Probability of disruption = 0.30
Probability of normal condition:
$$ 1-0.30=0.70 $$
Therefore:
$$ EU(R1)=(0.70)(70)+(0.30)(10) $$$$ =49+3 $$$$ \boxed{EU(R1)=52} $$

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R2 — City Road
Probability of disruption = 0.10
Probability of normal condition:
$$ 1-0.10=0.90 $$$$ EU(R2)=(0.90)(55)+(0.10)(20) $$$$ =49.5+2 $$$$ \boxed{EU(R2)=51.5} $$

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R3 — Ring Road
R3 has no disruption:
$$ EU(R3)=1(50) $$$$ \boxed{EU(R3)=50} $$
Comparison

| Route | Expected Utility |
|---|---|
| R1 – Highway | 52 |
| R2 – City Road | 51.5 |
| R3 – Ring Road | 50 |

Therefore, the rational agent chooses the route with the highest expected utility:
$$ \boxed{\text{Choose R1 – Highway}} $$

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(c) New Radio Alert — 4 Marks
The radio alert says there is a major accident on the highway.
Probability of disruption on R1 becomes:
$$ P(D)=0.80 $$

Therefore:
$$ P(\text{Normal})=1-0.80=0.20 $$
The utilities remain:

- Normal R1 = 70
- Disrupted R1 = 10

Therefore:
$$ EU(R1)=(0.20)(70)+(0.80)(10) $$$$ =14+8 $$$$ \boxed{EU(R1)=22} $$
The other routes remain:
$$ EU(R2)=51.5 $$$$ EU(R3)=50 $$
Therefore, after receiving the new percept:
$$ 51.5>50>22 $$
So the rational agent changes its decision to:
$$ \boxed{\text{Choose R2 – City Road}} $$
Important point
The physical roads have not changed, but the agent's information about the probability of disruption has changed. Therefore, its expected utilities and rational decision change.

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(d) Actual Disruption and Rationality — 4 Marks
In part (b), the rational agent chose R1 because:
$$ EU(R1)=52 $$
was the highest expected utility.
Suppose the accident actually occurs.
Then R1 takes:
$$ T=70 $$
and the taxi arrives after the deadline:
$$ L=1 $$
Therefore, actual utility is:
$$ U=100-70-20(1) $$$$ \boxed{U=10} $$
Was the agent irrational?
No.
The agent was rational because it selected R1 based on the information and probabilities available before the trip. An accident was possible with probability 0.30, but it was not certain.
A rational agent chooses the action that maximizes expected performance, not the action that guarantees the best outcome.
Rational Agent vs Omniscient Agent
Rational agent:

- Makes the best decision using the percepts and information available to it.
- Maximizes expected performance.
- Does not know future events with certainty.

Omniscient agent:

- Knows the actual future outcome in advance.
- Would know whether the accident would occur before choosing the route.

Therefore, an omniscient agent could avoid R1 if it knew beforehand that the accident would occur, but a normal rational agent cannot be expected to have that knowledge.
$$ \boxed{\text{Actual poor outcome does not necessarily mean the decision was irrational.}} $$

(e) Interpretation — 3 Marks
Rationality depends on the percept sequence and the performance measure.
The performance measure here rewards shorter travel time and gives an additional penalty for arriving after 55 minutes. Therefore, the agent must consider both travel time and the probability of being late.
In part (b), the available percepts gave:
$$ P(\text{R1 disruption})=0.30 $$
so R1 had the highest expected utility:
$$ EU(R1)=52 $$
In part (c), the new radio alert provides additional information:
$$ P(\text{R1 disruption})=0.80 $$
This changes the expected utility:
$$ EU(R1)=22 $$
Therefore, R2 becomes the rational choice.
Key interpretation
Rationality is based on the information available to the agent at the time of decision-making. A new percept can change the estimated outcomes and therefore change the rational action, even when the physical environment itself has not changed.

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Final Answers
$$ \boxed{R1:\ EU=52} $$$$ \boxed{R2:\ EU=51.5} $$$$ \boxed{R3:\ EU=50} $$
Part (b):
$$ \boxed{\text{Choose R1}} $$
After radio alert:
$$ \boxed{EU(R1)=22} $$$$ \boxed{\text{Choose R2}} $$
If R1's accident occurs:
$$ \boxed{U_{\text{actual}}=10} $$
but
$$ \boxed{\text{the original choice of R1 was still rational.}} $$

### Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary (100% of marks) | Level 3 – Proficient (75%) | Level 2 – Developing (50%) | Level 1 – Beginning (25%) |
|---|---|---|---|---|---|
| **Problem Understanding** | **20%** | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| **Method Selection** | **20%** | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| **Solution Process** | **25%** | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| **Accuracy of Calculation** | **20%** | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| **Interpretation of Results** | **15%** | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |

| Q. No. / Criteria | Problem Understanding (4) | Method Selection (4) | Solution Process (5) | Accuracy of Calculation (4) | Interpretation of Results (3) | Sub. Total (20) | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |
| 4 | | | | | | | |
| Total | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |